Working with Layers in NVIDIA Omniverse

A practical guide for factory simulation and digital twin development

1. What Are Layers?

Layers in Omniverse are based on USD (Universal Scene Description). Think of them like Photoshop layers for a 3D scene — each layer is a separate USD file that composes into one unified scene. You can edit, hide, or swap layers independently without affecting the rest of the scene.

Factory simulation analogy:

- Root Layer = the factory building and floor
- Sublayer 1 = current equipment layout
- Sublayer 2 = proposed layout changes (what-if scenario)
- Toggle Sublayer 2 off to compare before/after without losing any work

2. Types of Layers

Layer Type	What It Is	Factory Simulation Example
Root Layer	The base USD file — the foundation of your scene. All sublayers compose on top of it.	Factory building shell, floor, walls
Sublayer	An additional USD file stacked on the root. Can be toggled on/off independently.	Equipment layout, shelving racks, conveyor belts
Session Layer	Temporary edits stored in memory only. Lost when Omniverse is closed. Always at the top of the stack.	Quick test moves, temporary positioning
Authoring Layer	Whichever layer is currently active for editing. Only one layer can be the authoring target at a time.	The layer you are actively placing equipment into

3. How to Open the Layers Panel

The Layers panel is not enabled by default in Kit Base Editor. Follow these steps to enable it:

1. Go to Developer menu in the top menu bar
2. Click Extension Manager (or press Ctrl+Shift+E)
3. In the search bar, type: layers
4. Find omni.kit.widget.layers and click Enable
5. Go to Window in the top menu bar
6. Click Layers — the panel will appear in your UI

Once open, the Layers panel shows the full layer stack for your current scene, including the Session Layer at the top and the Root Layer at the bottom.

4. Creating a Sublayer

Sublayers let you separate different parts of your factory layout into independent files. This is useful for comparing layout scenarios or collaborating with multiple engineers.

7. In the Layers panel, click the + button
8. The Create Sublayer dialog opens — navigate to your save location (e.g. Documents)
9. Type a name in the File name field (e.g. layout_v1)
10. Click Save
11. A prompt may appear: 'Root Layer is not empty. Transfer Root Layer contents to the new sublayer?' — click No to keep the root layer unchanged

Your new sublayer (layout_v1.usd) will now appear in the Layers panel beneath the Session Layer.

5. Setting the Authoring Layer

The Authoring Layer is the active layer that receives your edits. Only one layer can be the authoring target at a time. Any objects you create, move, or modify go into the authoring layer.

Switch to a sublayer as the authoring target:

12. In the Layers panel, right-click your sublayer (e.g. layout_v1.usd)
13. Click Set Authoring Layer
14. The layer will highlight — you are now editing into this sublayer
15. Add objects (e.g. Create > Mesh > Cube) — they are stored in layout_v1.usd only

Switch back to the Root Layer as the authoring target:

16. In the Layers panel, right-click the Root Layer
17. Click Set Authoring Layer
18. Edits now go into the root layer again

Important: You must switch the authoring layer back to the Root Layer before you can toggle a sublayer's visibility on or off. Omniverse will not let you mute (hide) the layer you are currently editing.

6. Toggling Layer Visibility

Toggling a sublayer's visibility on or off lets you compare different layout scenarios without deleting anything.

How to toggle a sublayer:

19. First, make sure the Root Layer is the authoring target (see Section 5)
20. In the Layers panel, click the eye icon next to your sublayer (e.g. layout_v1.usd)
21. The layer is now hidden — objects in that layer disappear from the viewport
22. Click the eye icon again to make the layer visible

Factory simulation use case: Toggle your 'proposed layout' sublayer on and off to show stakeholders the difference between the current and proposed equipment arrangement — without touching the base scene.

7. Common Issues and Fixes

Issue	Cause	Fix
Cannot mute authoring target	You are trying to hide the layer you are currently editing	Switch authoring layer to Root Layer first, then toggle the sublayer eye icon
Edits going into wrong layer	Authoring layer is not set to the intended sublayer	Right-click the correct layer in the Layers panel and click Set Authoring Layer
Layers panel not visible	Extension not enabled	Developer > Extension Manager > search 'layers' > enable omni.kit.widget.layers
Changes lost after closing	Edits were in the Session Layer	Always set a named sublayer as the authoring target before making changes

8. Quick Reference

Action	How To Do It
Open Layers panel	Extension Manager > enable omni.kit.widget.layers > Window > Layers
Create a sublayer	Layers panel > click + > name file > Save
Set authoring layer	Right-click layer in Layers panel > Set Authoring Layer
Toggle layer visibility	Switch to Root as authoring target first, then click eye icon on sublayer
Switch back to Root	Right-click Root Layer > Set Authoring Layer